Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/13/2005
PI Name:	Peterson, Kevin Ph.D.		
Project Title:	Using Molecular Clocks And Regulatory Networks To Test Origin-Of-Bilateria Scenarios		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1584		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Because of the wealth of both phylogenetic and developmental data, there is now a tremendous interest in the origin of bilaterian metazoans and the associated paleontological events such as the Cambrian explosion. It is now generally accepted that the latest common ancestor of bilaterians was a fairly complex animal complete with anterior-posterior and dorsal-ventral axes, and a central nervous system. Nonetheless, it is still unclear if the origin of this latest common ancestor shortly preceded the Cambrian explosion, as suggested by the fossil record, or if there was a significant but cryptic period of bilaterian evolution before their first appearance in the fossil record, as suggested by most molecular-clock studies. Understanding which stage of the bilaterian life cycle is primitive for Bilateria, the larval stage or the adult stage, is crucial for deciphering the temporal incongruence between the fossil record and these molecular-clock analyses. To test for homology of larval types, we cloned a gene in the purple sea urchin		

Task Description:	Strongylocentrotus purpuratus called NK2.1, a member of the NK-2 class of homeodomain transcription factors. NK2.1 is expressed at the apical pole of the sea urchin, and transcripts are eventually restricted to the larval apical organ. Because the apical organ is restricted to primary larval forms, homology of the apical organ implies homology of the larva. In order to test this homology, we want to test for the conservation of gene batteries underlying apical pole development in sea urchins and in gastropod molluscs. We have already identified several important upstream factors in the sea urchin, and are in the process of examining these same genes in the mollusc. Determining the polarity of the larval vs. adult bilaterian body plan will then allow for a better understanding of major events in the history of life on Earth including the Cambrian explosion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ascertaining the primitive life cycle of bilaterians will shed tremendous insight into the ecological ramifications of surviving catastrophic environmental perturbations such as "snowball earth". Indeed, current discussions among snowball-earth workers have focused, in part, on the inferred ecology of the animals surviving the glaciation(s) (see Runnegar 2000, and discussion by Schrag & Hoffman 2001 and Hyde et al. 2001). Thus, a proper understanding of the earliest stages of bilaterian evolution is necessary in order to properly understand the geological and paleontological events of the Late Neoproterozoic. Finally, given that bilaterians may have been present for ~ 100 Ma before they made their first appearance in the fossil record, small but potentially complex metazoans may have thrived and diversified without leaving a trace on our own planet in a relatively recent time frame. Discussions centered on discovering complex life throughout the universe should keep this most interesting possibility in mind.
Task Progress:	During the last year, we have placed NK2.1 into the gene regulatory network governing apical tuft development. NK2.1 is downstream of the transcription factor HNF-6, and is upstream of many cilia-type genes including radial spoke 3, radial spoke head, dynien light chain p33, and alpha-tubulin. Abrogation of NK2.1 activity abolishes development of the apical tuft, and results in embryos that can not swim properly. In addition, we have isolated all of these genes from the gastropod mollusc, and the initial expression patterns are very different: HNf6- and the few cilia genes we have examined are not co-localized. This is consistent with our hypothesis that the dipleurula and trochophore larval forms evolved independently. Both manuscripts (sea urchin and gastropod) should be submitted by the end of the summer.
Bibliography Type:	Description: (Last Updated: 02/23/2007)
Articles in Peer-reviewed Journals	Peterson, K.J. & Butterfield, N.J. "Origin of the Eumetazoa – testing ecological predictions of molecular clocks against the Proterozoic fossil record." Proceedings of the National Academy of Sciences, USA, in press. , Jul-2005
Articles in Peer-reviewed Journals	Higgs, H.N. and Peterson, K.J. "Phylogenetic analysis of the forming homology 2 domain. " Molecular Biology of the Cell 16: 1-13. , Jan-2005
Articles in Peer-reviewed Journals	Peterson, K.J., McPeek, M.A., and Evan, D.A.D. "Tempo and mode of early animal evolution: inferences from rocks, Hox, and molecular clocks. " Paleobiology, in press , Jul-2005